

3. Vacuum Cleaner Problem

Aim

To simulate a Vacuum Cleaner Agent.

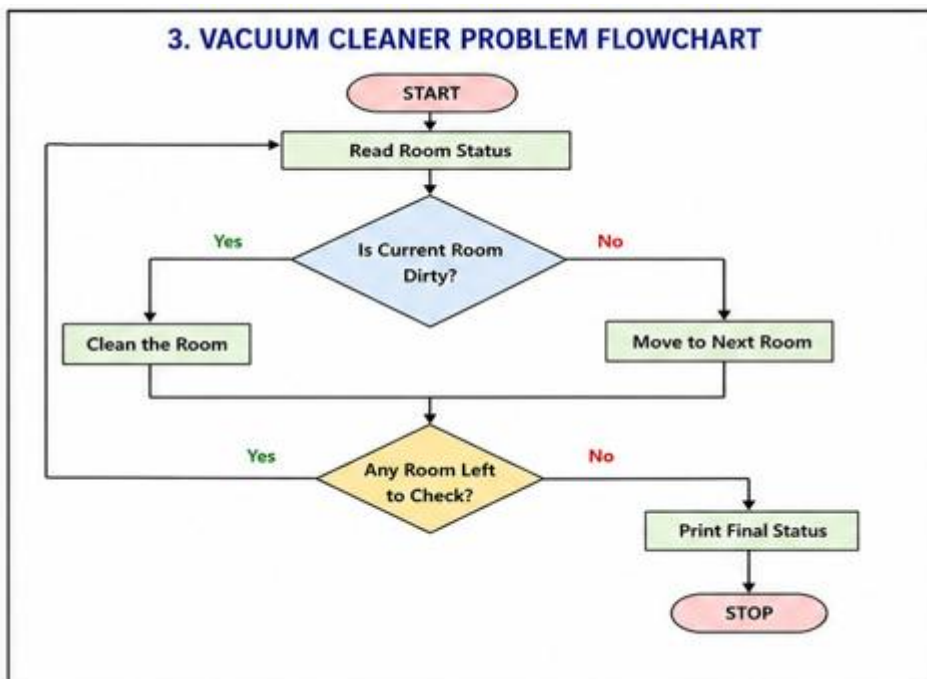
Objective

To clean all dirty rooms.

Algorithm

1. Check current location.
2. If dirty, clean.
3. Move to next room.
4. Repeat until all rooms are clean.

Flowchart



Python code:

```
rooms = {}
```

```
n = int(input("Enter number of rooms: "))
```

```
for i in range(n):
```

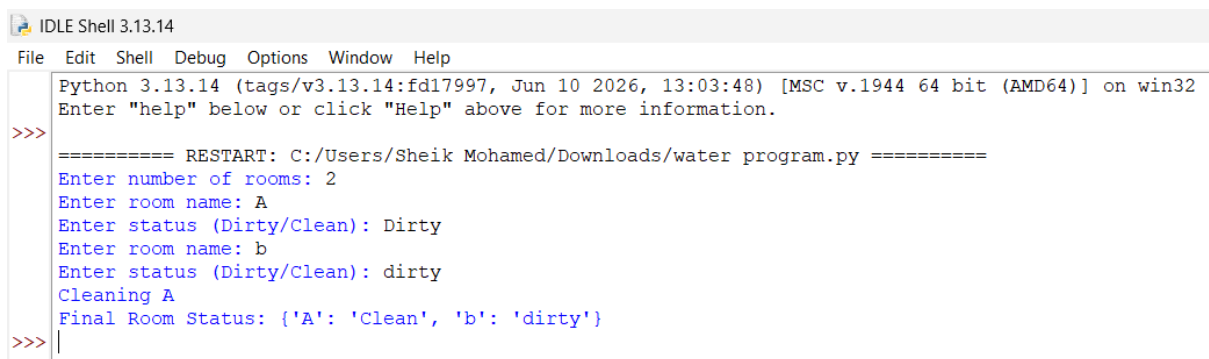
```
room = input("Enter room name: ")
status = input("Enter status (Dirty/Clean): ")
rooms[room] = status
```

for room in rooms:

```
    if rooms[room] == "Dirty":
        print("Cleaning", room)
        rooms[room] = "Clean"
```

```
print("Final Room Status:", rooms)
```

output:



```
IDLE Shell 3.13.14
File Edit Shell Debug Options Window Help
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/Sheik Mohamed/Downloads/water program.py =====
Enter number of rooms: 2
Enter room name: A
Enter status (Dirty/Clean): Dirty
Enter room name: b
Enter status (Dirty/Clean): dirty
Cleaning A
Final Room Status: {'A': 'Clean', 'b': 'dirty'}
>>>
```

Result

All rooms were cleaned.

Conclusion

The Vacuum Cleaner Agent successfully achieved its goal.